

DIRECTIONS

Show your work in the box for each problem.

Write your final answer on the answer line.

Use the strategies from Unit 3: column add and subtract with regrouping, cascade through zeros, and estimation.

Name: _____

Date: _____

1 Column addition with regrouping

Stack and add. Show every regroup.

$$3,875 + 3,249 = \underline{\hspace{2cm}}$$

SHOW YOUR WORK

Sum: _____

2 Column subtraction with regrouping

Stack and subtract. Show every unbundle.

$$7,123 - 1,456 = \underline{\hspace{2cm}}$$

SHOW YOUR WORK

Difference: _____

3 Cascade through zeros

Stack and subtract. Cascade the borrow through zeros.

$$8,000 - 3,765 = \underline{\hspace{2cm}}$$

SHOW WHAT EACH ZERO BECAME

Difference: _____

4 Estimate by rounding (3-digit)

Estimate by rounding each number to the nearest hundred.

$$895 + 756 \approx \underline{\hspace{2cm}}$$

SHOW YOUR ROUNDING

Estimate: _____

5 Estimate by rounding (4-digit)

Estimate by rounding each number to the nearest hundred.

$$4,223 + 758 \approx \underline{\hspace{2cm}}$$

SHOW YOUR ROUNDING

Estimate: _____

6 Visual/modeling — draw the chart

Draw a place-value chart for $9,344 + 466$.

Add column by column, marking each regroup with a small 1 at the top of the next column.

DRAW YOUR CHART AND LABEL EVERY REGROUP

Sum: _____

7 Two-step word problem

A 4-seat sofa set costs \$1,309. The 3-seat sofa set costs \$810 less than the 4-seat. How much do both sofa sets cost altogether? Show both steps.

STEP 1 (3-SEAT PRICE) + STEP 2 (BOTH TOGETHER)

Total cost: _____

8 Find the mistake

A student wrote: ' $4,206 - 335 = 3,971$. I unbundled in the tens column.'
Is the student right? If not, what is the correct answer and what did they do wrong?

EXPLAIN THE MISTAKE

Correct answer: _____

9 Reasoning — which estimate is more precise?

For $4,223 + 758$, two students estimate two ways:

Student A rounds each to the nearest THOUSAND. Student B rounds each to the nearest HUNDRED.

The actual answer is 4,981. Whose estimate will be MORE PRECISE? Explain.

WRITE YOUR REASONING

More precise: _____

10 Cascade word problem

Aki started with 5,057 beads. After her last project she has 398 beads left.

How many beads did Aki USE on her last project? Show your work with the cascade.

SHOW YOUR STACK WITH THE CASCADE

Beads used: _____